Ericsson PE Node BAS-PE-10 Basavanagudi POP Optical Transport Failure Resolution SOP

Node: BAS-PE-10 | **Vendor:** Ericsson | **Severity:** Major | **Alarm Code:** ALARM_4777

Description

The failure of the Optical Transport subsystem in the Ericsson PE Node BAS-PE-10 located at the Basavanagudi POP can be attributed to the degradation of the SFP/XFP transceivers, fiber cuts, or Rx/Tx power loss due to the inherent limitations of the optical transmission technology. The Ericsson Node B is equipped with the SFP/XFP slots that support various optical interfaces, including GPON, EPON, and XGPON. The optical transmission is governed by the Optical Transport Network (OTN) protocol, which operates at the physical layer. The OTN protocol ensures the reliable transmission of optical signals over fiber-optic cables. However, the SFP/XFP transceivers and the fiber-optic cables are prone to degradation and physical damage, leading to signal attenuation and power loss. Furthermore, the Ericsson Node B has multiple SFP/XFP slots, and the failure of one slot does not necessarily affect the entire system, as the redundant links and SFP/XFP slots are designed to provide high availability. Nevertheless, the failure of a critical SFP/XFP slot or a fiber cut can lead to a significant impact on the network performance and availability. In this SOP, we will guide you through the troubleshooting and resolution of the Optical Transport failure in the Ericsson PE Node BAS-PE-10, focusing on the SFP/XFP degradation, fiber cuts, and Rx/Tx power loss.

Pre-requisite Checks

- Verify that the Basavanagudi POP is accessible and that the remote hands are dispatched to the site.
- Check the redundant link status and ensure that it is operational.
- Verify that the Ericsson Node B is powered on and that all fans are operational.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

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Command: show sfp status, Expected Output: SFP status: OK/NOT OK, Description: This command displays the status of the SFP/XFP slot, indicating whether it is operational or not.

Command: show interface detail, Expected Output: Interface status: UP/DOWN, Rx/Tx power: OK/LOW, Description: This command displays the status of the interface, including the Rx/Tx power levels, to help
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identify any power loss issues.

Command: show log , Expected Output: Log messages related to SFP/XFP degradation or fiber cuts, Description: This command displays the log messages related to the SFP/XFP degradation or fiber cuts, providing valuable information for troubleshooting.
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Resolution Steps

Step 1: Step: Replace the faulty SFP/XFP transceiver, Command: replace sfp , Description: This command replaces the faulty SFP/XFP transceiver with a new one, resolving the SFP/XFP degradation issue.

Step 2: Step: Repair the fiber cut, Command: repair fiber , Description: This command repairs the fiber cut, restoring the optical signal and resolving the fiber cut issue.

Step 3: Step: Adjust the Rx/Tx power settings, Command: set interface rx-tx-power , Description: This command adjusts the Rx/Tx power settings, ensuring that the power levels are optimal and resolving the Rx/Tx power loss issue.

Post-Check Verification

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Command: show sfp status, Expected Output: SFP status: OK, Description: This command verifies that the SFP/XFP slot is operational after the resolution.

Command: show interface detail, Expected Output: Interface status: UP, Rx/Tx power: OK, Description: This command verifies that the interface is operational and that the Rx/Tx power levels are optimal after the resolution.

Command: show log , Expected Output: No log messages related to SFP/XFP degradation or fiber cuts, Description: This command verifies that there are no log messages related to the SFP/XFP degradation or fiber cuts after the resolution.
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Roll-back Procedures

- Step: Restore the original SFP/XFP transceiver, Command: restore sfp , Description: This command restores the original SFP/XFP transceiver, reverting the changes made during the resolution.

- Step: Repair the original fiber, Command: repair fiber , Description: This command repairs the original fiber, restoring the optical signal and reverting the changes made during the resolution.

- Step: Reset the Rx/Tx power settings to the original values, Command: reset interface rx-tx-power, Description: This command resets the Rx/Tx power settings to the original values, reverting the changes made during the resolution.